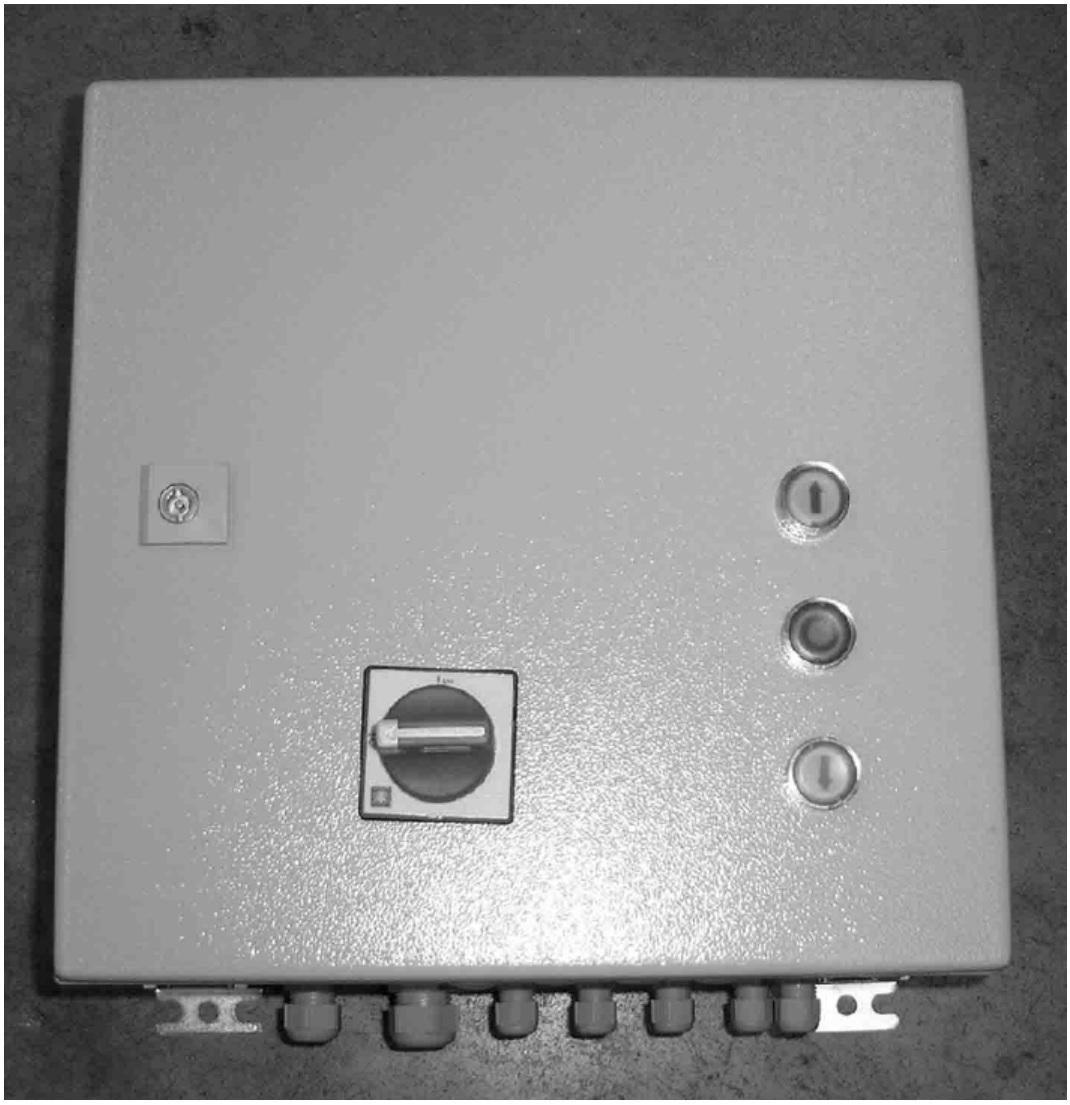
Control ACS 60



6410R0038

8 Setup operation manual

8.1 Turning on the frequency converter drive ACS 60

8.1.1 Preparation / Information

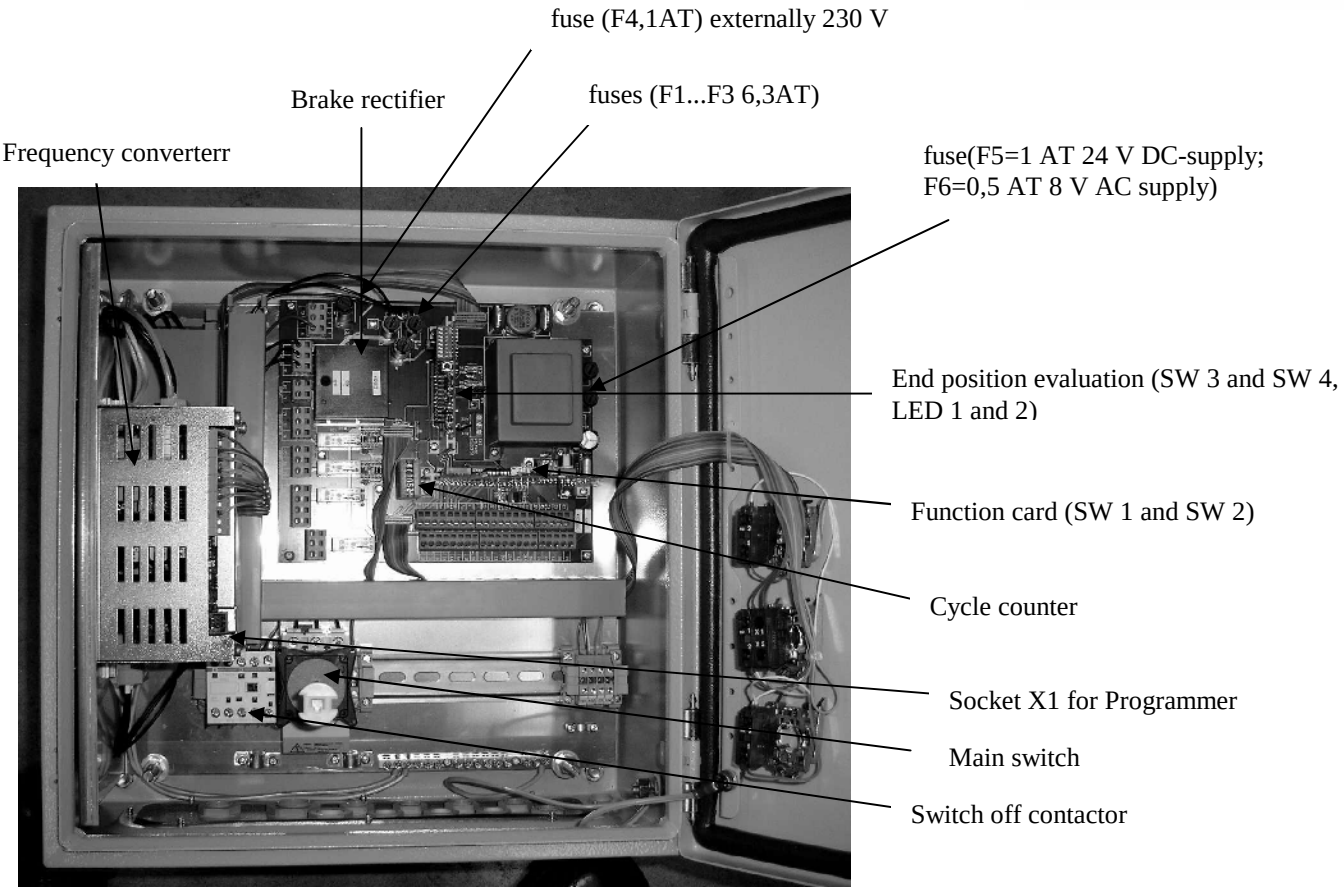
Before starting the set-up operation of the FU drive the following requirements must be fulfilled:

- Make sure that there will be no traffic of persons or cars within the vicinity of the door during set-up operation.
- The door is installed according the installation manual.
- All connecting power lines between door and drive are installed.
- All external impulse activators and safety features are connected.
- Unused inputs are bridged according the wiring diagram.
- If a FI-safety switch is used it must be an all-power-sensitive type (type B). Only this type controls alternating currents and pulsing DC fault current. However, this safety switch should not be the only safety feature for indirect touch (VDE 0160). In case of doubt please get advice from a skilled electrician. The door must be connected to the equipotential bonding.
- The connection cable (option) is not plugged in or the main switch is turned off !
  The supply line should not be less than 1,5 mm2 in diameter. The prefuses must be selected according the conductor cross section and selectivity to the prefuses of the drive (T6,3A). The power supply must only be done by qualified electrician.
- The door should be half open.

The connection cables should be installed according the following drawings. The cables should be put in a right angle. The power supply lines should be separately installed from the drive lines, otherwise it could lead to malfunctions. Put plugs with respective protection category on unused inputs.



Connecting the shielded cables (motor, brake and incremental shaft encoder cable) please pay utmost attention on good large-surface contact of the shielding with the mounting base. Otherwise there may be errors in function of the door !



picture 1: control panel inside

For the connection of the shield, the coating of the cable must be removed. Do not damage the shield! The now open metal shield is fixed with the screwed-on clip at the mounting base. The unshielded cable between shield clip and connection pin should be as short as possible.

To avoid unauthorized changes of the control function a special programming device is necessary for the set-up operation and the changes of the door parameters on the frequency converter. The plug of the programming device must be plugged into the socket (X1) of the frequency converter. Correct procedure is described in 8.2.

8.1.2 Description of the mentioned safety symbols

In this operation manual the following safety symbols are used. These symbols are meant to notice the text for safety advice.

These symbols advice on existing dangers for life and health of persons or machines, material or the environment.



=> Attention! Danger for persons or door system



=> Attention! Danger by electrical power

This symbol supplies information for a better understanding of the operation of the door processes.



=> Important information

Pay utmost attention to place the advice directly at the door and the control panel. The advice must be kept well readable at all times.

8.1.3 Installation of the drive unit

The installation of the control panel is done with the 4 pre-assembled fixing brackets in vertical position. Fix at the wall with 4 hexagon wood screws 8 x 60 mm (DIN 571) and dowel 10 mm. On steel construction the fixing can be done with threaded screws M8 x 30 mm. Perhaps fixing must be done according local situation. The drive must be installed in oscillation- and vibration free position.

8.2 Set-up procedure

8.2.1 Selection of position actuator

To operate a door it is necessary to determine the position of the door. Two possibilities are applicable.

a.) The door is equipped with a position actuator. This actuator (incremental shaft encoder) receives impulses from the motor. These impulses are being counted and afterwards the door position may be determined. The reference limit switch (end position door open) determines this way the zero point.

b.) The door is equipped with 4 free adjustable limit switches. The limit switches incorporate the following functions:
- door open,
- Start of upper brake deceleration ramp,
- Start of bottom brake deceleration ramp,
- door closed
- By activating the limit switches the door position is determined.

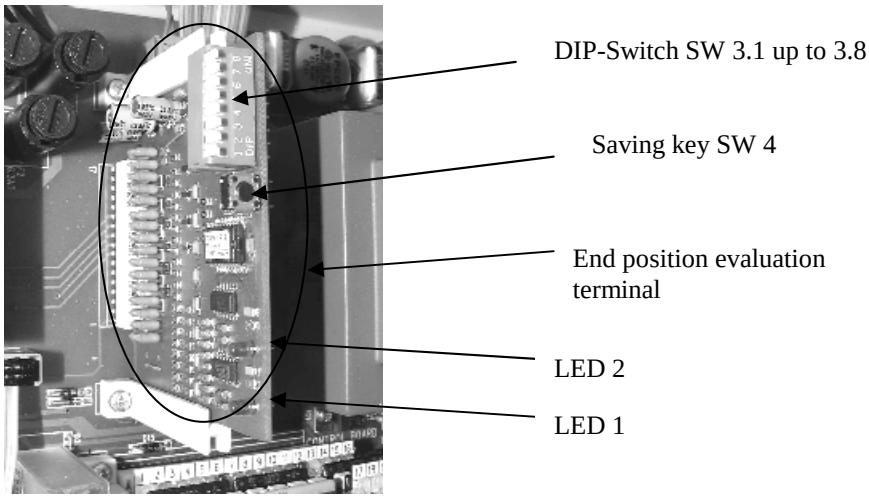


The limit switches of the brake deceleration ramps must strictly remain activated until reaching the end positions!

The selected procedure of the door must be adjusted at the position activator terminal (see pict. 2) at the DIP switch SW 3.2:

SW 3.2 = ON => limit switch version (right switch position)
SW 3.2 = OFF => position actuator version (left switch position)

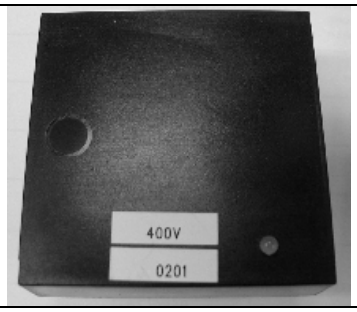
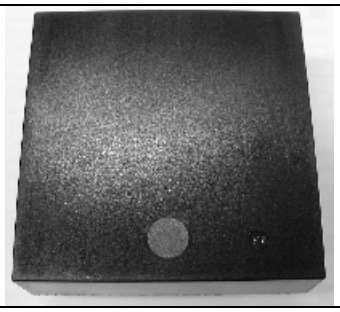
This adjustment may only be changed in 'Power Off' condition !
Upon wrong adjustment damages may occur to the door !



picture 2: end position evaluation terminal

8.2.2 Brake voltage

The brake voltage is generated via an installed pluggable rectifier (see picture 3). The brake rectifier must be chosen according the supply voltage of the brake. Through the color of the installed diode the different types can be selected.

		
400V power/ 196 V brake voltage green LED (one-way rectifier)	230V power / 196 V brake voltage red LED (bridge rectifier)	Relay module (external brake voltage) without LED

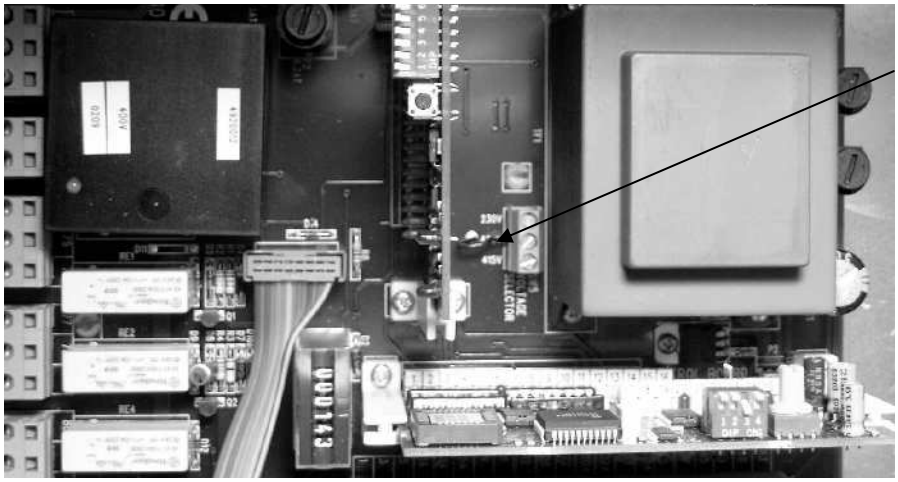
picture 3: Overview rectifier / brake relay

The drive cannot be operated on 230 V AC, when powered by the frequency converter. This means the bridge rectifier cannot be used with this drive !

If there is an external rectifier incorporated within the motor, the relay module can be plugged into the drive. The switch contact of the external rectifier is connected to the terminal (70/71) in the drive. The relay switches the DC current of the external rectifier.

The frequency controlled power of the motor cannot be used for power supply of the brake rectifier. Do NOT connect the external brake rectifier to the motor line, but feed it via external AC supply !

8.2.3 Selecting the power supply voltage



-

picture 4: wire strap with terminal block

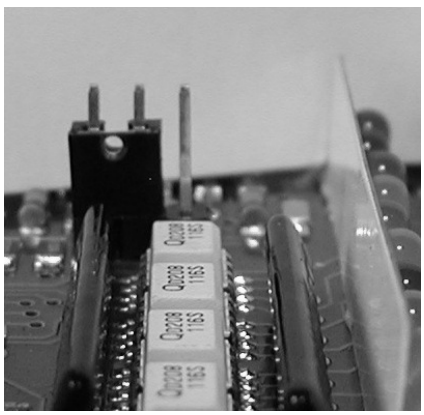
At the main terminal is a terminal block including wire strap where the correct voltage can be selected for the control system.
This adjustment must only be 400 V and 415 V AC for both areas.

Due to using the Frequency converter, do NOT select 230 VAC.

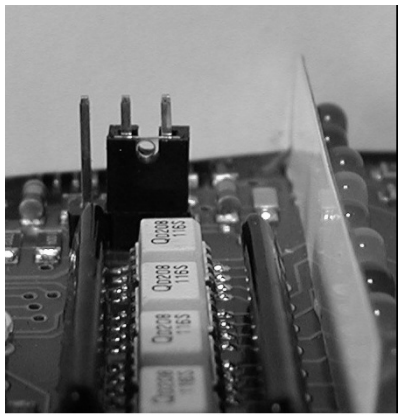
8.2.4 Selection of the closing safety edge

The drive can be operated with 2 different types of closing safety edge.
- resistor evaluation 8,2 kΩ
- pneumatic safety edge

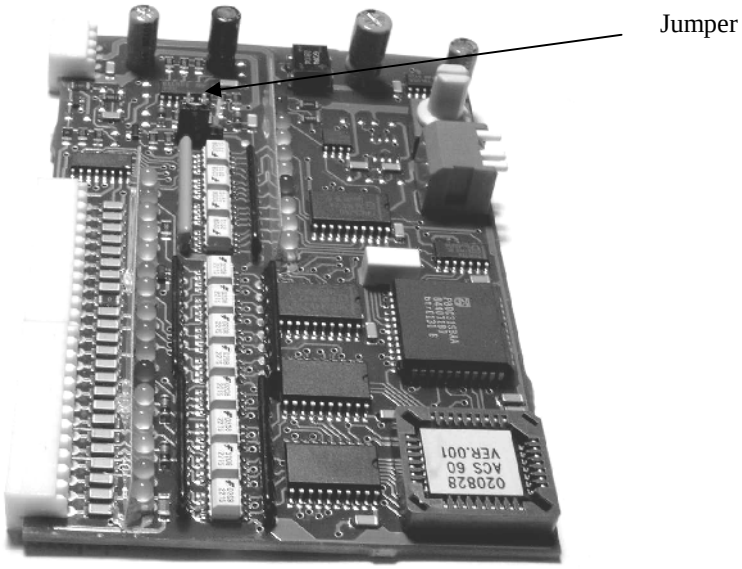
The selection of the respective safety edge is done via the jumper on the function card. (see pictures 5, 6 and 7).



picture 5: elektric safety edge with 8,2 kΩ resistor



picture 6: pneumatic safety edge



picture 7: Function card

If the jumper bridges both contacts facing the terminal block (picture 5) the resistor evaluation is selected. LED 16 is off. If both contacts facing the DIP-switch SW 2 are bridged (pict. 6), pneumatic evaluation is selected. LED 16 is lit.

8.2.5 Activating the programming mode

This adjustment must only be activated for complete set-up operation if the drive has been turned off!

To reach the programming mode the DIP switch SW 3.1 must be turned to ON.

8.2.6 Programming the frequency converter

Before the first operation of the drive the converter must be programmed. This means all door specific values, like frequencies, ramps, etc. must be activated in the converter. Therefore the connection cable of the programming unit (see pict. 8) must be plugged onto plug X1 (Western plug) of the frequency converter with Power OFF. The drive must then be turned ON via the power plug (option) or via the main switch.
After that, one of the following text messages shows at the operating display. By activating the arrow keys you may jump through the menu. The key with the arrow pointing downwards leads to the next point of the menu, the key with the upward arrow returns to the former point of menu. With the key „+"- or „-„ the values may be adjusted. By activating the „+" key you go to the next value, with the „-„ key you return to the former value. Following is the selection of the possible text messages available.

The parameters of the door type shown are automatically activated. With the function „Paraload" the saved values from the operation part are transferred to the converter!



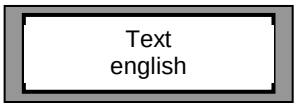
Text display

Arrow keys to menu selection

„+"- and „-„ key for adjustment of values

picture 8: converter programming unit

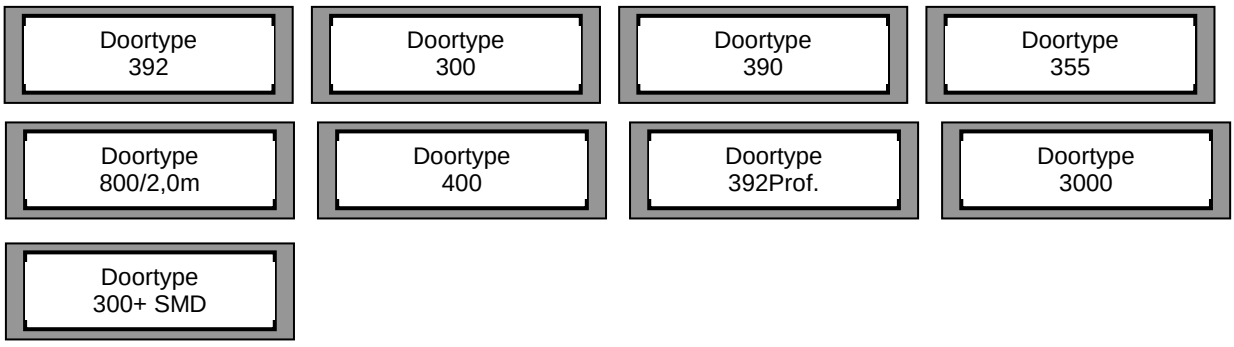
8.2.6.1 Display language



By activating the „+" or „-„ key the languages German, English, French and Spanish can be chosen. The standard text language is German.

8.2.6.2 Door type selection

By activating one of the arrow keys one of the following door types is showing.



For each door type special parameters for each individual door type are structured! Depending on the used software version also deviating door types can be shown.

The door types 5 to 20 are reserved for further, not yet used door types and are currently without any function. The door type special 1 is reserved for a special door type and must not be adjusted !
By activating the „+" or „-„ key the selected door type may be adjusted.

The parameters of the shown door type are automatically activated !

With the function



the preset values (at manufacturers site) which are saved in the operation unit can be transferred to the converter.



This function may only be implemented if the parameters at the operation unit have current status! In case of doubt please contact the technical sales department for help.

The parameters in the converter are activated by pushing and keeping hold of the „+“ key.
During activation the following text message shows



if the text

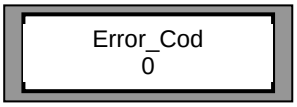


shows, the parameters in the converter are activated and the „+“ key does not need to be activated again. After error free activation the following text message shows



8.2.6.3 Error code message

Errors in the converter can be read in the display of menu point



Here the shown number signals the following error:
The message „0“ means that there is no error at the converter.
Following error codes are possible:

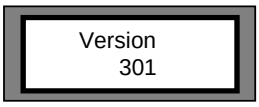
Code	malfunction	cause
1	Under-Voltage	Voltage at FU (intermediate circuit) is to low
2	Excess voltage	Input voltage or intermediate circuit voltage is to high

4	Excess temperature	temperature of FU is to high. Vicinity temperature is higher than 40 C°
7	Over-Voltage	max. voltage has been exceeded
8	malfunction EEPROM	EEPROM at FU is defect or has error
9	Peak voltage	max. peak voltage has been exceeded
16	No release	at 20 channel terminal of FU between clamp 4 (0 V DC) and clamp 10 (release) voltage applied must be bigger than 13 V
64	Short circuit/short circuit to ground	motor supply cable, motor or FU has short circuit or short circuit to ground, motor was heavily overloaded
250	Reset via Watch Dog	Internal malfunctions in program procedure, drive is automatically reset

Resetting error is only possible by turning off the main switch or by pulling the CEE plug (option) off.

8.2.6.4 Indication of software version

In the following window the applied software version used by the converter Eprom is shown.



301 shows the software in use. The actual shown version may differ from the one described here because of applied changes.

The plug of the programming unit can be unlatched by pushing the catch lever a little bit and it may then be taken off the converter.

8.2.7 Adjustment of upper end position



There will be no automatic limitation of the upper end position. Limitation only via limit switch end position „door open“.

Incremental shaft encoder version: (SW3.1 = on, SW3.2 = off)
Before the first run the reference limit switch must be adjusted ! After activation of the open key at the control panel the door runs in hold to run mode in creeping speed up to the reference limit switch. There the door stops automatically. A closing action is not possible at this point of the menu. A zero action run must be performed after every turning on the drive. This position does not need any saving. The upper end position must always be contained on the reference limit switch.

Limit switch version: (SW3.1 = on, SW3.2 = on)
At the limit switch version the limit switch for the upper end position must be adjusted directly to the end position. In order to determine this position easily the door can be run to the upper end position in the hold to run mode. Now the limit switch can be adjusted correctly.

A zero action run must be performed after every turning on of the drive. With it the door runs automatically up to the end position “door open” (creeping speed) after an opening impulse by the key at the control panel door.

8.2.8 Adjusting upper brake ramp



A brake distance that has been selected to short may lead to an overrun of the end position !

The starting point of the upper brake ramp must be adjusted below the upper end position. The following table can be used for rough determination of the distance (see table 1).

Door type	300	390	355	400	392	RT 800 (2,0 m)	300+ SMD
Upper brake distance in mm (measured from bottom level rubber profile)	560 mm	440 mm	500 mm	830 mm	550 mm	530 mm	600 mm

Table 1: standard value for upper brake distance

With reaching this point the frequency converter starts to decelerate the motor from high frequency to a low one (creep speed run). If this distance is set to short the door runs with to much speed to the end position „door open“ and cannot decelerate the speed fast enough. The end position „door open“ is overrun ! Should the distance be set to long, it may still be optimized after set-up operation.



The optimum brake distance is applied if the door decelerates from high into low frequency in a flowing action and proceeds to the end position in creeping speed up to a full stop.



For easier adjustment the single brake distance positions may be marked at the column (i.e. adhesive strips). This will make it easier to correct the positions.

To set the upper brake distance (door open) the respective door position can be reached in hold to run mode via the open and close key at the control panel door.

Incremental shaft encoder version:
Thereafter the DIP switch SW 3.5 must be set to ON and the saving key should be briefly activated. After saving action the DIP switch must be set to OFF position again. This position is now saved.

Limit switch version:
Upon reaching the starting point of the upper brake ramp the limit switch must be activated. For easier adjustment the door should run in creeping speed in the set-up operation mode (hold to run mode).Adjustment can be made via the open and close key at the control panel door.

The limit switch for the upper brake ramp (door open) must always remain active until reaching the upper end position !

8.2.9 Adjustment of bottom brake ramp

The drive has a technical specific after-running distance. Please make sure to keep the following values. A selected brake distance being too short may lead to overrun of the end position !

The starting point of the bottom brake ramp must be set above the bottom end position. The following table may be used as standard value for the distance (see table 2).

Door type	300	390	355	400	392	392 Prof.	RT 800 (2,0 m)	300+ SMD
Brake distance bottom in mm (measured from bottom edge of rubber profile)	320 mm	180 mm	190 mm	270 mm	180 mm	180 mm	140 mm	160 mm

Table 2: Standard value for bottom brake distance (door closed)

At reaching this point the frequency converter starts to decelerate the motor from high to low frequency (creeping speed). If this distance is set to short the door runs with to much speed into the bottom end position and cannot decelerate sufficiently. The bottom end position will be overrun ! Should the distance be set to long it may be optimized after set-up operation.

The optimum brake distance is reached if the door decelerates from high to low frequency in a flowing movement up to a full stop.

For setting this point the door can be moved to the respective position via the open and close key at the control panel door.

Incremental shaft encoder version:
Thereafter the DIP switch SW 3.6 must be set to ON and the saving key should be briefly activated. After saving procedure the DIP switch must be set to OFF position again. This position is now saved
.

limit switch version:
Upon reaching the starting point of the bottom brake ramp the limit switch must be activated. For easier adjustment the door should run in creeping speed in the set-up operation mode (hold to run mode). The adjustment may be done via the open and close key at the control panel door.

8.2.10 Adjustment of bottom end position



During set-up operation there is no limitation to the bottom end position. The door may be run in hold to run operation over the bottom end position !
The drive has a technically related brief after-running distance. Please, make sure to keep the following values.

Because of the after-running of the drive the door must not be completely closed during saving of this position. As a rough standard value for the after running distance of the door the following table may be used (see table 3). This means during saving of this position the door must at least be positioned around the below mentioned distance above the later end position, i.e. at 300 door at least 90 mm!

Door type	300	390	355	400	392	392 Prof.	RT 800 (2,0 m)	300+ SMD
After-running distance for bottom end position in mm (measured from bottom edge of rubber profile)	90 mm	130 mm	180 mm	160 mm	130 mm	130 mm	130 mm	70 mm

Table 3: After-running distance bottom end position (door closed)

By activating the close key at the control panel door it is possible to run the door in hold to run mode to the bottom end position. After reaching this position the DIP switch SW 3.7 must be set to ON and the end position should be saved with the saving key (see pict.1). Thereafter the DIP switch must be set to OFF position again. This position is now saved.
In the following table (see table 4) the function of the single DIP switches of the end position value terminal is shown in an overview.

DIP switch	Position	Function	Position	Function
3.1	on	program mode	off	Operation mode
3.2	on	Limit switch evaluation	off	Incremental shaft encoder evaluation
3.3	on	No function	off	No function
3.4	on	No function	off	No function
3.5	on	Upper Brake ramp	off	No function
3.6	on	Bottom brake ramp	off	No function
3.7	on	Bottom end posi-tion	off	No function
3.8	on	No function	off	No function

Table 4: Function DIP switch of the end position evaluation terminal

8.2.11 Finishing set-up operation mode

After adjustment of all positions the set-up operation mode must be deactivated. For that the DIP switch SW 3.1 is set on OFF position. Please check if the DIP switch 3.3 to 3.8 is also set on OFF position. The adjustment of SW 3.2 must remain set in accordance to the incremental shaft encoder procedure. Now it is still possible to correct the saved adjustments (see 8.2.10). If this is not necessary, the kind of operation for the drive must be selected. (see chapter 8.3)

8.2.12 Correction of the end position and the brake ramp



Before correction of the different end positions the door should always be run up to the reference limit switch

To correct the bottom end position or both brake ramps the DIP switch 3.1 must be set to ON (Programming mode). Thereafter the position to be corrected can be approached. After turning on the respective DIP switch, for example SW3.6 for the bottom brake ramp (door close), this position can be saved individually with the saving key. After saving the changed position the respective DIP switch must be set on OFF position again. To finish the programming mode the DIP switch 3.1 must be set on OFF position again.



If correcting the values please consider the after running distances!

8.2.13 Error message of the incremental shaft encoder terminal

Both LED´s at the end position terminal (LED 1and 2) have the following meaning (table 5).

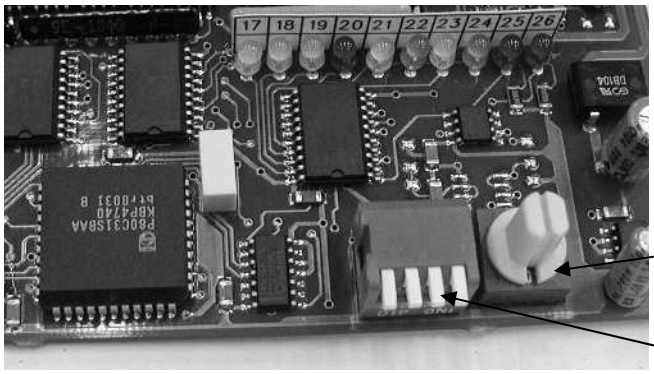
LED 1	LED 2	display
Off	flashing	Error set-up operation (no positions has been saved
Off	On	Error in direction (channel A and B of the incremental shaft encoder are mixed up)
On	On	Error incremental shaft encoder (no impulses are received at the input of the drive)
flashing	flashing	Error reference limit switch (reference limit switch is defect or switch does not get any release at certain time after door movement)
flashing	Off	Set-up operation mode (reference run necessary)
On	Off	Set-up operation mode (drive requests position)
On	flashing	Wrong incremental shaft encoding procedure selected (Incremental shaft encoder/limit switch) or changeover to limit switch is faulty
flashing	On	no function
Off	Off	Normal function mode without error

Table 5: Error message of the end position evaluation terminal

To confirm the error the drive must be turned off and on again. Because of the big internal capacitors, at converter error the drive must only be switched on again after about 10 seconds. This time is necessary for the converter to discharge the capacitors.

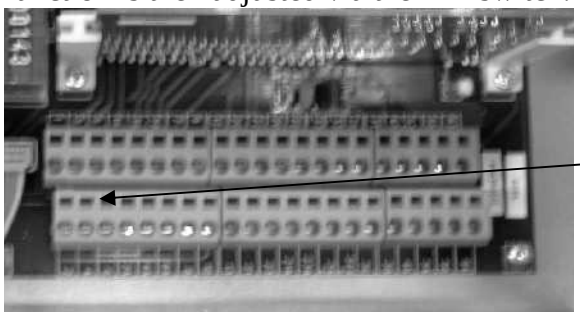
8.3 Adjusting the type of operation

The type of operation of the drive is adjusted via the DIP switch SW1.1 to .4 at the function card. In picture 8 the DIP switch SW1 and the hexadecimal coded switch SW2 is shown.



picture 8: DIP switch SW1 and 2

From the following table (see table 6) the type of operation may be selected with the respective switch positions. The respective impulse activators are connected at the pins 3/4, 3a/4a and 5/6, 5a/6a. The function is then adjusted via the DIP switch.



picture 9: terminal block

SW1.1	SW 1.2	SW 2	pin 3/4	pin 5/6
ON	ON	0	Toggle circuit	Toggle circuit
ON	ON	1-F	Park house function + closing after time	Toggle circuit
ON	OFF	0	open	Toggle circuit
ON	OFF	1-F	open + close after time	Toggle circuit
OFF	ON	0	Park house function	close
OFF	ON	1-F	open + close after time	open + close after time
OFF	OFF	0	open	close
OFF	OFF	1-F	open + close after time	close

Table 6: Adjustment of type of operation

The following function can be programmed by the two DIP switches SW1.3 and 1.4:

SW1.3 = Locking
With this switch it is determined whether the door should be locked in the position „door open“ after activation of safety edge or whether closing operation starts automatically after duration of keep open time (function „close after time“ must be additionally activated). Upon safety edge operation the adjustment should always be positioned at ON position (lock).

Signal lamp drive:
SW1.4: ON = Signals during the opening and closing run
The exit relay (RE4) for the signal lamp drive is activated if an opening impulse is activated at closed door. It remains activated until the end position „door open“ is reached. After duration of the keep open time the prerunning time of the signal lamp of the drive starts and the relay is activated again. Prerunning time is fixed at 2 seconds, thereafter the closing movement of the door starts automatically. The signal lamp relay remains activated until the door is completely closed.

SW1.4:OFF = Signal only during closing movement
At this variant the relay (RE4) closes at the moment after duration of keep open time and the prerunning time starts. After duration of prerunning time (2 seconds) the closing operation starts. The relay remains activated as long as the end position „door closed“ has been reached.
Maximum load of the relay contacts is at 250V AC / 8A. The contacts are available as locking contacts. The exact electrical connection may be taken from the enclosed wiring diagram. This wiring diagram is located at the folder inside the control panel.

8.3.1 Description of the types of operation

Close after time

Opening of the door is done via manual or automatic impulse activator. After duration of keep open time the door closes automatically. The keep open time only starts if the open impulse is activated and all safety features are free. The keep open time is adjusted at the hexadecimal coded switch SW 2 (see picture 8). This function may also be activated in connection with the park house function.

Toggle circuit

Opening of the door is done via manual or automatic impulse activator. However, closing operation follows only at the next impulse activation. There is no automatic closing after time.

Park house function:

Opening of the door is done via manual or automatic impulse activator. Thereafter the door closes automatically 2 seconds after the safety 2 (stationary photocell) has been activated and is free again. If safety 2 is not activated, the door remains open ! Additionally to the park house function a function „closing after time“ can be activated. This means if the park house function could not be activated, the door closes automatically after duration of preset keep open time. The adjustment of the keep open time is done via the hexadecimal coded switch SW2. (see picture 8)

8.3.2 Keep open time

The keep open time for the automatic closing after time is adjusted via the hexadecimal coded switch SW2. This time may be varied between minimum 1 second (adjustment SW2 to 1) and maximum 60 seconds (SW2 to F). For exact time adjustment see table 7. But if SW2 is set on 0 the door remains

permanently open after an opening impulse. There will be no automatic closing after time. Only a closing impulse activates the closing movement.

SW2	Automatic closing after keep open time of:
0	no automatic closing
1	1 sec.
2	2 sec.
3	3 sec.
4	4 sec.
5	6 sec.
6	7 sec.
7	10 sec.
8	13 sec.
9	16 sec.
A	19 sec.
B	22 sec.
C	25 sec.
D	30 sec.
E	45 sec.
F	60 sec.

Table 7: time for automatic closingAdditional functions

Ready for operation/ error message

The error / ready for operation relay (RE5) becomes active after turning on the drive upon error free function and it remains active until turning off if there is not been an error. In case of error it becomes deactivated for signaling error. This relay is furnished with a potential free change-over-contact. Take the exact electrical connection from the enclosed wiring diagram. The wiring diagram is located in the respective folder inside the control panel. The maximum load to the relay contact is 250V AC / 8A.

Messages door open/close

The messages „door open“ (RE1) and „door closed“ (RE2) are available each as a potential free change-over-contact. The connection to the relay is done via plug-type connection terminal. Take the exact electrical connection from the enclosed wiring diagram. This wiring diagram is located in a folder inside the control panel. The maximum load for the relay contact is 250V AC / 8A. With a limit switch version and activation of the stop button the relays for the end positions become inactive!

Supply of external assembly units

Via the pins T3 / N external assembly units (i.e. traffic lights, flashing alarm lamp, etc.) may be supplied with 230 V AC power. This voltage is secured with fuse F4 (1 AT). For that the N-conductor of the input to the drive must be connected.

8.4 Pin configuration of drive

Configuration of connection pins at the main terminal is as follows:

T1,T2,T3: input voltage
T3,N: supply voltage for external assembly units (only possible with connection of N-conductor)
2, 4, 6: Motor supply at off contactor (fail safe relay)

70,71:	Brake
1,2,1A,2A:	Stop inputs
3,4,3A,4A:	inputs external impulse activators
5,6,5A,6A:	inputs external impulse activators
7,8:	Safety edge
12,10:	limit switch door open
9,10	reference limit switch
12,13:	limit switch door closed
22,22A:	+24V DC
20,20A:	0V DC
21,21A:	+24V DC with test
23:	Safety 1
23A:	Safety 2
35,36:	Temperature contact
37,38:	Emergency lever
45,46:	Crash sensor (opener)
53,54:	no function
55,56:	reserve exit
34,34A:	+24V DC
33,33A:	0 DC
63,64:	24V AC

Incremental shaft encoder connections / limit switch connections:

101	24 V DC
102	0 V DC
103	channel A/ pre-limit switch brake distance „door closed“
104	channel B/ pre-limit switch brake distance „door open“

Potential free messages may be taken from the following connections:

74,75,76:	door open (74=COM, 75=NO, 76=NC)
77,78,79:	door closed (77=COM, 78=NO, 79=NC)
80,80A,81,81A:	Signal light drive (80,80A=COM, 81,81A=NO)
82,83,84:	malfunction/error (82=COM, 83=NO, 84=NC)

The wiring for the individual door type must be made according the enclosed wiring diagram.

8.5 Meaning of the diodes

The arrangement, function and assignment of the diodes of the function card is shown in table 8.

LED-no.	Function	color	description
1	Open key	yellow	Lights up on active external open impulse
2	Stop key	green	Off, if stop has been activated
3	Close key	yellow	Lights up on active close impulse
4	Upper limit switch	yellow	Lights up at activated limit switch „door open“

5	Bottom limit switch	yellow	Lights up at activated limit switch „door closed“
6	safety edge	red	Lights up at released safety edge
7	photocell 1	green	off, if safety has been released
8	Temperature switch	green	off, if temperature switch is activated
9	Emergency lever	green	off, if emergency lever is activated
10	Open key at control panel	yellow	Lights up if open key at the control panel is activated
11	Crash sensor (opener)	green	off, if crash is activated
12	Programming mode	yellow	off, if drive is in programming mode, lights up at operation mode
13	Test close relay	yellow	on during creeping speed run
14	Close key at control panel	yellow	Lights up, if close key at the control panel is activated
15	photocell 2	green	off, if safety is activated
16	Pneumatic safety edge	yellow	Off, if safety edge is activated
17	Open run	yellow	Lights up during open run
18	Close run	yellow	Lights up during close run
19	Brake	yellow	Lights up if brake is activated (electr. released)
20	Error message	red	Lights up at malfunctions/error
21	Traffic light control	yellow	Lights up if traffic light is on
22	Ready for operation message	green	Lights up if drive is ready for operation
23	Reserve exit	yellow	Lights up
24	Test of safety devices	green	Goes briefly off during the test of safeties, otherwise permanently on.
25	Error message photocell	red	Lights up on faulty photocell
26	Error message safety edge	red	Lights up on faulty safety edge

Table 8 Assignment of diode

9 Operation Manual

9.1 General Information frequency converter drive ACS 60

9.1.1 Introduction

The ACS 60 drive has been developed for the use with industrial door systems. The drive complies with the valid CE regulations.

The drive contains of all equipment parts necessary to operate doors frequency converter driven under consideration of all safety criteria. The characteristic of the frequency converter drive ensures a soft running start and deceleration of the door system. This incorporates the advantage that for one thing the mechanical part of the door is saved from hard crash incidents and on the other hand the door can be operated at a high speed level.

The frequency converter and the drive terminal is located at the control panel together with the function card and the incremental shaft encoder terminal. At the main terminal are additional relays for realization of potential free contacts for the door status messages and special functions.

9.1.2 General danger information

Safety features of the door are designed for ordinary operation with vehicles according EN 12453 regulation norm, resp. ZH 1/494 regulation.

Additional safety devices or impulse activators may become necessary under consideration of local conditions, especially with pedestrian traffic.

For questions and information, please contact your local Albany Door Systems Agent.

9.2 Basic design

9.2.1 Control cabinet with operation elements

The door drive is installed on a fixing plate in a steel metal box with measurements 380mm x 380mm x 210mm. The standard lock is a turnbuckle with a two-way key (other locks upon request). In the bottom is a connecting plate for the arrangement of the connecting cables to the door as well as to the external impulse activators. The power supply cable is also guided through this fixing plate. The inner side of the control panel door has a storage bag for the wiring diagram, door operation manual, etc. The protection category of the control panel is IP55.

The main switch (option) and 3 push buttons for operating the drive (see pict. 2) are built into the door of the control panel. The drive is switched on and off with the red main switch which at the same time is the EMERGENCY OFF switch (option). The main switch can be locked in the switched off position via a shackle type connector. At standard drives the CEE-plug (CEEkon-plug) serves as power supply device. At the left hand side of the top hat rail are the connecting pins for the supply line. The power supply line is already connected with the CEE-plug (CEEkon-plug) on these pins at the factory site. At the main switch version the supply line is already connected directly to the pins (L1, L2, L3, (N), PE) at factory site. The N-conductor is only needed if external components shall be operated on 230 V.



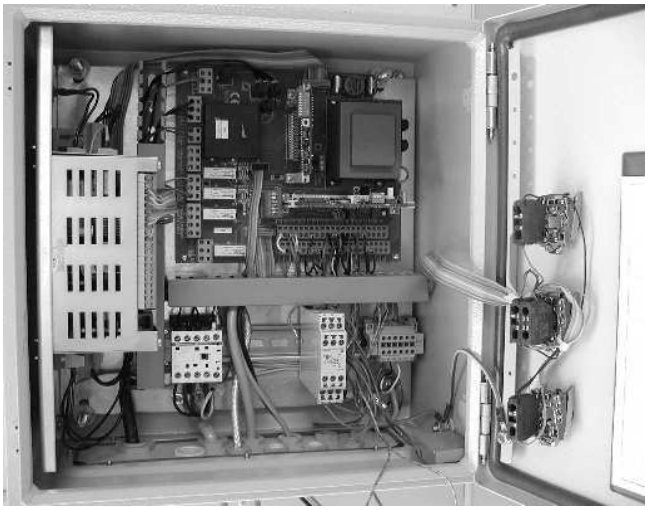


Abb. 1: control panel outside **Abb. 2: control panel inside**

The three push buttons „open(reset), stop and close“ are made for the operation of the door and to reset the drive in case of malfunction. Optionally the open button may also be made as permanent open button.
Located within the control panel is a fixing plate to which the drive and the frequency converter are installed. The fixing plate serves at the same time as cooling device for the final position of the frequency converter.

Due to the cooling device function of the fixing plate, the plate can store high temperatures during operation. Danger for burns !

The input filter is installed behind the converter. Between the drive and the converter is the cable conduit for the connecting lines. In the bottom area of the fixing plate is a hat top rail screwed onto where the pins are positioned for the incremental shaft encoder, an off contactor and the pins for the motor line. Furthermore it is also possible to connect for example radio receivers, induction loop detectors or elements for special functions. Below the hat top rail is a PE rail installed for the connection of the earth conductor. The shield of the motor, the brake and the incremental shaft encoder lines must be tightly connected with the fixing plate via clamps.

In the low area of the control panel a hat top rail is installed to which additional devices may be connected like induction loop detectors, radio receivers, etc.

The CEE-plug (CEEkon-plug) serves at the same time as power turn off device !
Only qualified electricians are allowed to work on the electric system ! The connection of the drive to the supply voltage must be made according the valid regulations !

The cable guidance is done from below through the screwed cable gland at the control panel.

9.2.2 Drive terminal and frequency converter

The whole drive comprises a compact designed printed board (function and end position evaluation terminal are plugged onto) and a separately positioned frequency converter. On the drive the following elements are installed:
- short circuit proof power pack for the supply of the micro processor drive,

- the micro processor with Eprom and the end position evaluation,

- pull spring connection clamps.

The frequency converter is installed at the left hand side beside the drive on the fixing plate. At the output of the converter an additional off contactor is connected already at factory site. On this contactor the direct connection to the motor (clamps 2, 4, 6, PE) must be made. On top of the converter at the terminal X1 the connection line to the drive must be connected. The green LED on the converter lights permanently at regular operation and flashes upon malfunction.



Through the intermediate circuit capacitors the converter still keeps high voltage –up to 3 minutes after switching off the drive-.

9.2.3 Connection positions for the drive

The connection positions for the drive are described in detail in the set-up operation manual of the ACS 60.

9.2.4 Function of the operation keys

The three keys at the control panel door serve as follows:

„OPEN“ key
The upper key serves as „OPEN“ key. This key gives the OPEN order to the drive.

„STOP“ and „RESET“ key
The key in the middle serves as „RESET“ and at the same time as „STOP“ key. With this key the door can be stopped. At the same time the door gets locked. The door is set into operation again after the next open or close impulse. The red signal light in the „STOP“ key serves as error signal (see 9.7.4). Malfunctions can be reset with the „RESET“ key.

„CLOSE“ key
With this key a closing impulse is given to the drive. Since after every close impulse the safety devices are tested firstly, the closing movement starts only after a short time delay.

9.3 Function

The positioning of the door can be done via an incremental shaft encoder or through four free adjustable limit switches. With both versions a zero run must take place after turn on. Furthermore, a reference run must be made with an incremental shaft encoder version after an activation of the emergency lever, so that the drive can find it’s zero point again (see 9.3.1).

9.3.1 Switch on

Plug in the power plug or switch on the main switch (option). After a short time the red „Stop“ key is flashing. The lock of the door is released by activating the “Reset“ resp. „Open“ key. After that a reference run is made automatically.

Should the door be completely open during switch on operation the “Open” key must be activated once.

After duration of keep open time or after a „Close“ impulse the door closes and is ready for operation.

9.3.2 Open run of the door

An „Open“ impulse may be given by any impulse activator or at the key board. The brake is released, the drive motor is driven by the frequency converter according preset acceleration ramp. The door opens with maximum speed up to the starting point of the decelerating ramp. (Adjustable at the set-up operation menu). Now the door decelerates to about 1/6 of the opening speed through the frequency converter preset brake ramp. The door moves with creeping speed up to the preset upper end position. Upon reaching this position the motor is turned off and the brake is active.

Depending on the preset function, the keep open time starts now or the drive unit waits for a closing impulse. The safeties are active.

9.3.3 Close run of the door

After duration of the keep open time resp. at a closing impulse the connected safeties are tested at first by the control unit. If the safety test was successful the safeties are active, the brake is being released and the motor will be driven by the frequency converter according the preset acceleration ramp. The door closes with about 2/5 of the maximum speed up to the starting point of the deceleration ramp (Adjustable at set-up installation menu).

Now the door decelerates to about 1/6 of the opening speed according preset brake ramp. The door runs to the bottom end position in creeping speed. Upon reaching this end position the motor is being turned off and the brake is active.



About 50 mm before reaching the bottom end position the safeties are turned off.

9.3.4 Traffic light function

A further relay is planned for the signal light control. Two variants for signaling can be adjusted via the DIP switch SW 1-4. At the first variant SW 1-4 ON the relay gets activated during the opening movement and with a prerunning time of 2 seconds with the closing movement of the door. At the second variant SW 1-4 OFF the relay only switches during the closing movement with a prerunning time of 2 seconds.

DIP switch	position	function
SW 1-4	ON	Traffic light function during opening movement, traffic light function during closing movement with 2 seconds prerunning time
SW 1-4	OFF	Traffic light function during closing time with 2 seconds prerunning time

9.4 Safety features (see also chapter 9.7)

To avoid accidents the closing movement of an automatically running door must be observed. For this reason the following safety features are incorporated in the door.

Before each closing movement of the door the safety photocells are tested by the micro processor by turning them briefly off and on again. During this action the processor evaluates the number of inputs and compares the number which must be the same. Any deviation to the set value means a defect of the photocell and the closing movement does not start. The door can only be closed in “hold to run” operation. The testing is shown by the automatic brief elapse of the two LED´s (7 and 15) which show the safety inputs at the control unit.



In the hold to run operation all the safety devices are without function. The door moves as long as the key at the control panel is being pushed. Due to the automatic after-run the door cannot be completely closed because of the disabled safety devices since there is no after-run movement! Upon reaching the set upper- or bottom-end position, the door is automatically turned off.

9.4.1 Stationary safety photocell

The stationary safety photocell is installed in the lower area of the door. Also here the proper function is monitored by the control unit. If the photocell is activated at open door a reset operation follows and a renewed start of the keep open time after release of the photocell. An activation during the closing movement causes an immediate stop of the door with a following reopening run up to the reference limit switch (reversing). In this end position the door closes automatically after duration of keep open time and is not locked as with the safety edge.

9.4.2 Safety edge (only with door type RR300, 390, 400 or optionally)

The safety edge serves as closing edge safety. It comprises a rubber profile which is installed at the main closing edge of the door. A deformation of the rubber profile activates an electric impulse. The closing movement of the door is immediately stopped because of this impulse. Afterwards the door runs up again until reaching the reference limit switch position and is locked there.. Locking operation is shown by the red light of the stop key. Furthermore are the diodes 20 (error message) and 26 (error message safety edge) on. Activation of the „Stop“ key releases the lock again. After that the door is ready for operation again.

To avoid malfunction or failure of this safety device the control unit is checked before each closing movement. An interruption of the safety edge as well as broken cable, short circuit or big deviation of the resistance value from the nominal value is acknowledged by the control unit and is shown by a red light at the control panel. If there is an error at the safety edge the door can only be operated via the „Open“ and „Close“ key at the control box in “hold to run” mode. Then the door runs in creeping speed during the closing movement and in normal opening speed during the opening movement.

9.4.3 Pre-running photocell (only at door type RR 350, 355, 450 and RT 800)

The pre-running photocell runs about 100 mm below the bottom profile. It joins the open and closing movements of the door. In the lower end position of the door the pre-running photocell is being retracted. Activation of the pre-running photocell has the same effect as the activation of the stationary safety photocell. The pre-running photocell is turned off in this area since in the lower end position of the door a compulsory activation of the bottom profile is performed. The photocell is also being

checked for error free function before each closing movement of the door. The photocell is turned off as soon as it hits the bottom.

9.4.4 „Stop“ key

By activating the „Stop“ key the door may be stopped in every position. A following „Open“ or „Close“ impulse brings the door into normal function again.

9.4.5 Running time observation

The running time of the opening and closing movements are monitored in order to secure a safe automatic function of the door. If the respective end position is not being reached within the preset values after an opening or closing impulse, the control unit turns off the motor and the red error lamp at the control panel door lights up. At the same time the LED 20 lights up on the function card. Only after activating the „Reset“ key a renewed impulse puts the door into operation again.

9.5 Power failure

At a power cut the door stops immediately still in it’s current position. After return of power it is only ready for operation again after activating the „Reset“ key and a following reference run (see 9.3).

9.6 Emergency manual operation

At power loss, repair and maintenance work may become necessary to open the door manually.



Before activating the emergency lever always turn off the main switch or pull out the power supply plug. After activation of the emergency lever the door can partly open or close by itself !

Through the emergency lever at the side column the brake of the gear motor can be released via the bowden cable. Depending on the power of the counterbalance the door leaf can partly open or close itself. Now you may grip below the rubber profile in the middle of the door and push the door leaf upwards. With roll doors the curtain rolls up on the top roll. At the RT 800 type the segments are pushed upwards. In order to reach regular function of the door again the emergency lever must be put into it’s original position again. After that the control unit may be turned on again. Activation of the „Reset“ key leads to the start of a zero run. After finishing the zero run the door is ready for operation again.

Should the emergency lever still be active when switching on the main switch, the red error lamp at the key board flashes and the diode 9 at the function card is off. This error message can only be confirmed after bringing the emergency lever into it’s original position again. Thereafter the door is ready for operation again.



The door is not designed for a regular manual operation.

9.7 Safety features

9.7.1 Temperature switch

For the safety of the drive motor three temperature switches are incorporated in it's coil. Upon activation of these switches the motor has become overheated and the drive turns off immediately. At the same time the red lamp at the control panel door and the LED 20 light up. Now it is necessary to switch off the control unit to get the motor cooled down first. Then should the door and the control unit be checked for possible mechanical resp. electrical faults. Only after the motor has cooled down the control unit may be switched on again. If there is no error anymore the control unit is automatically reset and regular operation may be taken up again.

9.7.2 Fuses

Three power fuses (F1...F3 6 AT) are incorporated in the control panel.

Additionally exists another fuse F4 (1 AT) for the external 230 V supply. The fuses F5 (1 AT) and F6 (0,5 AT) are for the internal 24 V DC and 8 V AC supply.

As replacement only fuses of the same type are to be used.

9.7.3 Crash sensor

With some door types it is possible to built in a knock out capability at the bottom profile in order to keep damages caused by hits as low as possible. If the door has been hit and the crash sensor is activated the door stops immediately. The error lamp at the control panel lights up. Now the disengaged bottom profile must be engaged again. This may be done in 2 different versions.

Door type RR 355, 450 :

At activated crash sensor the LED 11 is off. The door must now be completely closed in hold to run operation. For that the close key at the control panel must be held as long as the knock out capability has engaged again. A renewed „Open" impulse sets the door in operation again.

Door type RR 300 and 390:

The knock out capability unit must be manually engaged into the side column again. Thereafter the door will be ready for operation again after the first „open" or „close" impulse. At this variant the LED 2 is off upon activated crash sensor.



After each crash on the door it must be thoroughly checked for possible damages.

9.7.4 Error signal light and alarm relay

The integrated light in the stop key at the control panel signals together with the alarm relay different types of errors. During turn on operation the lamp flashes for 3 seconds to signal the turn on mode including it's self test operation. Faults occurring during this operation are shown immediately and the alarm relay is active. In switched off position of the control unit the alarm relay is also active. Further possible faults/errors with the respective messages are shown in the following table 4 (in any case the alarm relay is active).

Error light	Fault description
Off	Input voltage of control unit is missing
flashing	Switch on mode, for 3 sec. after switch on
flashes	LD9=Off=> emergency lever activated
On	Permanent impulse during the switch on mode at the control unit active
On	LD6=On=> error safety edge
On	LD25=On => short circuit in photo cell circle
On	LD7=Off => safety photocell interrupted
On	LD7=Off and control unit locked => temperature contact activated
On	LD26=On at SW1-3 ON => Safety edge activated
On	Control unit locked => maximum running time exceeded
On	Control unit locked => EPROM-Test resp. RAM-Test faulty

Tab. 1: Malfunctions

9.7.5 Error messages frequency converter

Errors within the converter may be read with the respective programming unit. Therein the current error code is shown. The exact meaning of the different error codes can be taken from the set-up operation manual, chapter 8.2.6.3. Also a detailed description of the exact procedure can be taken from chapter 8.2.